Vulnerability Assessment Report

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Executive Summary

This report details the findings of a vulnerability assessment conducted on TagBack on 30 March 2026. The purpose of this assessment was to identify security vulnerabilities that could be exploited by malicious actors and to provide recommendations for remediation. The assessment included network scanning and web application testing. Critical vulnerabilities were identified and immediate action is recommended.

Methodology

The following methodology was used during the vulnerability assessment:

1. **Information Gathering:** Collecting information about the target environment using both open-source intelligence (OSINT) and active reconnaissance techniques.
2. **Vulnerability Scanning:** Utilizing automated tools such as Nessus, OpenVAS and Burp Suite to identify potential vulnerabilities.
3. **Manual Testing:** Performing manual analysis to validate scan results and identify vulnerabilities that automated tools may miss. This included such as code review and penetration testing
4. **Exploitation (Optional):** In some cases, identified vulnerabilities were safely exploited to demonstrate the potential impact.
5. **Reporting:** Documenting the findings, including vulnerability descriptions, severity levels, and remediation recommendations.

Findings

The following vulnerabilities were identified during the assessment. Vulnerabilities are ranked by severity using the following scale:

- **Critical:** Immediate risk of compromise; requires immediate remediation.
- **High:** Significant risk; requires prompt remediation.
- **Medium:** Moderate risk; requires remediation in a timely manner.
- **Low:** Minor risk; requires remediation when resources are available.
- **Informational:** Not a vulnerability, but provides information about the system.

Details of each vulnerability, including a description, severity, affected component, and remediation recommendation, are provided below.

Vulnerability	Severity	Affected Component	Description	Recommendation
Unpatched Apache Server	Critical	Web Server	The Apache server is running an outdated version with known vulnerabilities.	Upgrade to the latest stable version of Apache.
Weak Password Policy	High	User Authentication	The password policy allows for weak and easily guessable passwords.	Enforce a strong password policy with minimum length, complexity, and password rotation.
Cross-Site Scripting (XSS)	Medium	Web Application	The web application is vulnerable to	Implement proper input validation and output

			XSS attacks.	encoding to prevent XSS.
Open Port 22 (SSH)	Low	Firewall	Port 22 is open to the internet, increasing the risk of brute-force attacks.	Restrict access to port 22 to only authorized IP addresses.

Recommendations

The following are general recommendations to improve the overall security posture:

- **Patch Management:** Implement a robust patch management process to ensure all systems are up-to-date with the latest security patches.
- **Hardening:** Harden systems according to industry best practices (e.g., CIS benchmarks).
- **Intrusion Detection:** Implement an intrusion detection system (IDS) to monitor for malicious activity.
- **Security Awareness Training:** Conduct regular security awareness training for all employees.
- **Regular Assessments:** Conduct regular vulnerability assessments and penetration tests to identify and address security vulnerabilities proactively.

Conclusion

This vulnerability assessment identified several security vulnerabilities that could potentially be exploited. Addressing these vulnerabilities is crucial to protecting the confidentiality, integrity, and availability of TagBack. It is recommended that the remediation steps outlined in this report be implemented as soon as possible, starting with the critical vulnerabilities. Further assessments should be conducted regularly to maintain a strong security posture. The assessment team is available to provide further assistance with remediation efforts.